NLP-BASED AIR QUALITY HEALTH ADVISORY GENERATOR

PROJECT REPORT

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ABSTRACT

Air pollution is an important environmental and public-health concern. Air-quality systems often present information as numerical values such as the Air Quality Index (AQI) and concentrations of pollutants. Although these values are useful, they may be difficult for non-technical users to interpret and connect to everyday decisions. This project proposes an NLP-based Air Quality Health Advisory Generator that converts Indian air-quality information into clear, actionable and personalized health guidance.

The system is designed around India's Central Pollution Control Board (CPCB) National Air Quality Index (NAQI) framework. It considers six air-quality categories—Good, Satisfactory, Moderate, Poor, Very Poor and Severe—and pollutant information for PM2.5, PM10, NO_2, SO_2, CO, O_3, NH_3 and Pb. The intended interface allows users to choose a city and user group, enter or adjust AQI and pollutant information, and provide weather context such as temperature, humidity and wind. The initial target cities are Bengaluru, Chennai, Delhi, Hyderabad and Mumbai.

The planned forecasting component uses a Long Short-Term Memory (LSTM) model to estimate future pollutant concentrations from time-series data. The advisory-generation component uses a Transformer-based large language model to express the analysed risk in natural language. Machine-learning models including LightGBM, XGBoost and Random Forest are also included in the proposed workflow for classification or supporting prediction tasks. Preliminary performance figures supplied for the project indicate approximately 89.9% accuracy for LightGBM, 89.8% for XGBoost and 87.7% for Random Forest; these figures should be interpreted in relation to the dataset and evaluation procedure used.

The expected outcome is a Streamlit web dashboard that helps users understand pollution levels and receive guidance appropriate to the selected population group. The advisories are intended to be consistent with CPCB categories and health-impact descriptions, and must be checked for accuracy, readability, relevance and safety before practical deployment.

Keywords: Natural Language Processing, Air Quality Index, CPCB, health advisory, LSTM, Transformer, machine learning, PM2.5, Streamlit.

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1. INTRODUCTION

Air quality changes across locations and time because of traffic, industrial activity, construction, domestic fuel use, weather and other environmental factors. Polluted air can affect the respiratory and cardiovascular systems. The risk depends not only on the overall pollution level but also on the pollutant involved, the duration of exposure and a person's susceptibility.

Air-quality monitoring platforms commonly report the AQI along with pollutant concentrations such as PM2.5, PM10, carbon monoxide (CO), nitrogen dioxide (NO_2), sulphur dioxide (SO_2) and ozone (O_3). These measurements are valuable, but raw numbers alone do not always tell a user what actions may be appropriate. A person may know that the AQI is high without understanding whether to reduce outdoor exercise, protect a child from prolonged exposure or take extra care when living with a respiratory condition.

Natural Language Processing (NLP) can bridge this communication gap by turning structured environmental measurements into concise text. In the proposed system, air-quality data are first validated and assessed against risk categories. A forecasting module estimates near-future pollutant levels, while a Transformer-based language model generates an advisory based on the structured results, selected population group and weather context.

The project focuses on Indian air-quality conditions and uses the CPCB National Air Quality Index categories as the basis for risk classification. The goal is not to replace official alerts or medical advice, but to make air-quality information easier to understand and act upon.

2. PROBLEM STATEMENT

Air pollution has a significant impact on human health, but air-quality information is often presented through technical values such as AQI, PM2.5, PM10, NO_2 and CO. Non-technical users may find it difficult to interpret these values or understand their health implications. Existing systems may provide a pollution reading or a generic warning without sufficiently adapting the message to the user's age group, sensitivity or local weather conditions.

Therefore, there is a need for a system that analyses AQI and multiple pollutant measurements, identifies the corresponding risk category, considers relevant environmental and user context, and generates clear, understandable and actionable health advisories using NLP.

3. RESEARCH GAPS

Numerical focus: Many systems emphasise AQI numbers and pollutant concentrations rather than explaining what the values mean for daily activities.

Technical presentation: Pollutant names and concentration values may be unfamiliar to users without an environmental or scientific background.

Generic warnings: A single warning may not address the different needs of children, older adults, people with asthma and the general population.

Limited natural-language generation: There is an opportunity to use NLP to convert structured measurements into coherent, readable advisories.

Multi-pollutant integration: An advisory should consider AQI and relevant pollutant measurements together instead of relying on a single value in isolation.

Forecasting and context: Historical pollutant patterns and weather conditions can help provide forward-looking context, although forecast uncertainty must be communicated.

Actionable guidance: The output should offer practical exposure-reduction steps and remain consistent with the selected risk category.

4. AIM AND OBJECTIVES

4.1 Aim

To develop an NLP-based system that converts Indian air-quality measurements and predicted pollutant trends into understandable, context-aware health advisories based on CPCB NAQI categories.

4.2 Objectives

- Analyse AQI and pollutant measurements, including PM2.5, PM10, NO_2, SO_2, CO, O_3, NH_3 and Pb where data are available.
- Classify air quality into the six CPCB categories: Good, Satisfactory, Moderate, Poor, Very Poor and Severe.
- Use historical time-series data with an LSTM model to forecast pollutant concentrations.
- Use a Transformer-based language model to generate clear natural-language health advisories from validated structured inputs.
- Personalize recommendations for children, elderly people, sensitive individuals, people with asthma and general users.
- Consider temperature, humidity and wind as contextual information when available.
- Provide a Streamlit dashboard for selecting a city, AQI, pollutant, weather conditions and user group.
- Evaluate the system for predictive performance, advisory relevance, readability, consistency and safety.

5. BACKGROUND AND CPCB NAQI FRAMEWORK

The project is intended to follow the Indian National Air Quality Index framework published by the Central Pollution Control Board. The supplied project specification identifies six categories and their broad health implications. The category mapping should be implemented using the official CPCB breakpoints and should be verified against the latest applicable CPCB documentation before release.

AQI category	AQI range supplied	Broad health implication
Good	0–50	Minimal health impact.
Satisfactory	51–100	Minor breathing discomfort may occur in sensitive individuals.
Moderate	101–200	Breathing discomfort may occur in people with lung or heart conditions.
Poor	201–300	Prolonged exposure may cause breathing discomfort to many

		people.
Very Poor	301–400	Prolonged exposure may cause respiratory illness; sensitive groups may be affected more severely.
Severe	401–500+	May affect healthy people and seriously affect people with existing health conditions.

Note: The supplied brief describes the Severe category as 401–500+. In implementation, the system should follow the exact current CPCB definition and handling of readings outside the published index range, rather than silently extending thresholds.

5.1 Pollutants considered

- PM2.5: Fine particulate matter that can penetrate deep into the lungs.
- PM10: Inhalable particulate matter associated with dust and other sources.
- NO_2: A gas commonly associated with combustion and traffic emissions.
- SO_2: A gas associated with certain industrial processes and fuel combustion.
- CO: Carbon monoxide, a gas produced by incomplete combustion.
- O_3: Ground-level ozone, which can irritate the respiratory system.
- NH_3: Ammonia, included in the pollutant set specified for this project.
- Pb: Lead, included in the pollutant set specified for this project.

The actual index calculation should use the official pollutant-specific breakpoints, averaging periods, units and data-validity rules. A missing pollutant measurement should not be treated as zero, and the system should identify when an advisory is based on incomplete data.

6. PROPOSED SYSTEM

The proposed system combines structured data processing, machine learning, time-series forecasting and NLP generation. Its purpose is to provide a readable explanation of air-quality risk rather than simply displaying a collection of values.

6.1 Input

Inputs may include city, current AQI, pollutant concentrations, dominant pollutant, temperature, humidity, wind speed or direction, forecast horizon and the intended user group. The initial city coverage is Bengaluru, Chennai, Delhi, Hyderabad and Mumbai. Data can be entered through the dashboard or supplied by a connected dataset or monitoring source, depending on the implementation.

6.2 Analysis

The system validates the inputs, checks units and missing values, derives the risk category using the CPCB-aligned classification logic, and identifies pollutants that contribute to the current condition. Where historical time-series data are available, the LSTM model estimates future pollutant concentrations.

6.3 NLP generation

The advisory-generation module receives a structured summary containing the city, AQI category, measured or predicted pollutant information, weather context, selected user group and approved recommendation rules. A Transformer-based language model then expresses this information in plain

language. The language model should not independently invent thresholds, diagnoses or unsupported medical claims; its output should be constrained by verified risk labels and recommendation templates.

6.4 Output

The dashboard displays the air-quality category, key pollutant, health-impact explanation and practical recommendations. Where a forecast is available, the interface should distinguish forecast values from observed measurements and indicate that predictions are estimates.

7. SYSTEM ARCHITECTURE AND METHODOLOGY

1. Air-quality and weather data collection: obtain current and historical records for the selected city or monitoring station.
2. Preprocessing: validate timestamps, units and ranges; handle missing or duplicate records; align pollutant and weather observations.
3. Feature preparation: construct time-series windows for LSTM forecasting and prepare tabular features for machine-learning models.
4. Risk classification: map AQI values to the six CPCB categories using approved thresholds and logic.
5. Forecasting: use historical pollutant concentrations and relevant features to estimate future concentrations.
6. Context analysis: identify the dominant pollutant, risk level, weather context and selected population group.
7. Advisory generation: provide the structured result to the Transformer-based language model and generate a readable recommendation.
8. Output validation: check that the advisory matches the category, includes appropriate precautions and does not contradict the structured analysis.
9. Dashboard presentation: show the result in the Streamlit interface.

7.1 Text-based workflow

Air-quality + weather data
↓
Preprocessing and validation
↓
CPCB-aligned risk classification + LSTM pollutant forecast
↓
Context and user-group analysis
↓
Transformer-based NLP advisory generation
↓
Rule and consistency checks
↓
Streamlit dashboard output

8. MACHINE LEARNING AND NLP COMPONENTS

8.1 Machine-learning classification models

The project specification lists LightGBM, XGBoost and Random Forest as candidate machine-learning models. These models can learn relationships between input features and a defined target, such as a risk class, when appropriately labelled training data are available. The supplied best reported accuracies are approximately 89.9% for LightGBM, 89.8% for XGBoost and 87.7% for Random Forest. The exact task, class distribution, data split and test procedure should be documented so the results can be interpreted correctly.

8.2 LSTM forecasting

Long Short-Term Memory (LSTM) is a recurrent neural-network architecture designed to learn patterns across sequences. Air pollution varies over time, and a time-series model can learn temporal dependencies from historical pollutant readings and associated features. For this project, an LSTM is planned to forecast pollutant concentrations. Forecasting performance should be evaluated on time-separated test data to reduce the risk of information leakage.

The forecast module should report its prediction horizon and error metrics. Suitable regression metrics include Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). A forecast must be presented as an estimate, not as a confirmed future measurement.

8.3 Transformer-based language model

A Transformer uses attention mechanisms to process relationships between tokens in a sequence. Transformer-based large language models can generate fluent, context-sensitive text from a structured prompt. In this project, the language model converts the system’s validated analysis into an advisory for the selected user group.

A recommended generation process is to supply the model with a fixed structure: city, observation time, AQI, CPCB category, pollutant readings, dominant pollutant, forecast status, weather context, user group and permitted action recommendations. The generated message should then be checked against rules to ensure that the category and precautions are consistent with the input.

8.4 Separation of responsibilities

Risk classification and pollutant threshold logic should remain deterministic and based on the CPCB framework. The LSTM is responsible for temporal prediction, while the Transformer-based language model is responsible for wording the explanation. This separation reduces the chance that a text-generation model changes an official category or fabricates a numerical threshold.

9. ADVISORY PERSONALIZATION AND EXAMPLE

9.1 Target user groups

General users: receive the broad exposure guidance appropriate to the detected category.

Children: receive cautious guidance about prolonged or strenuous outdoor activity when pollution is elevated.

Elderly people: receive additional exposure-reduction guidance when risk levels warrant it.

Sensitive individuals: receive extra precautions where respiratory or cardiovascular sensitivity is relevant.

People with asthma: receive careful exposure guidance and a reminder to follow their clinician-provided asthma action plan where applicable.

Personalization changes the emphasis and wording of the advisory; it must not change the measured AQI or the official risk category. The system should avoid diagnosing a condition or prescribing medication.

9.2 Example output

Example supplied in the project material: Delhi, AQI 150. The brief labels this condition "Moderately Polluted" and notes possible discomfort for people with lung or heart conditions, children and older adults. It identifies PM2.5 as the main pollutant and recommends reducing prolonged exertion and keeping windows closed if it is dusty.

Illustrative generated advisory: "Air quality in Delhi is moderately polluted. People with lung or heart conditions, children and older adults should reduce prolonged or strenuous outdoor activity. PM2.5 is the main pollutant in this example; fine particles can travel deep into the lungs. If outdoor air is dusty or pollution is elevated, reduce the entry of outdoor air where practical and follow local official guidance."

Implementation note: the CPCB category mapping in the supplied project brief identifies AQI 101–200 as Moderate, whereas the earlier PPT example uses the word "Unhealthy" for AQI 185. To maintain CPCB alignment, the application should use CPCB category names consistently rather than mixing them with another index's labels. The sample advisory should be generated from the final, verified CPCB category mapping.

9.3 Weather context

Temperature, humidity and wind are included as contextual inputs. These variables can help describe environmental conditions and may contribute to the forecasting model if supported by training data. They should not be used to arbitrarily change the CPCB AQI category. The system should clearly distinguish weather context from pollutant-based risk classification.

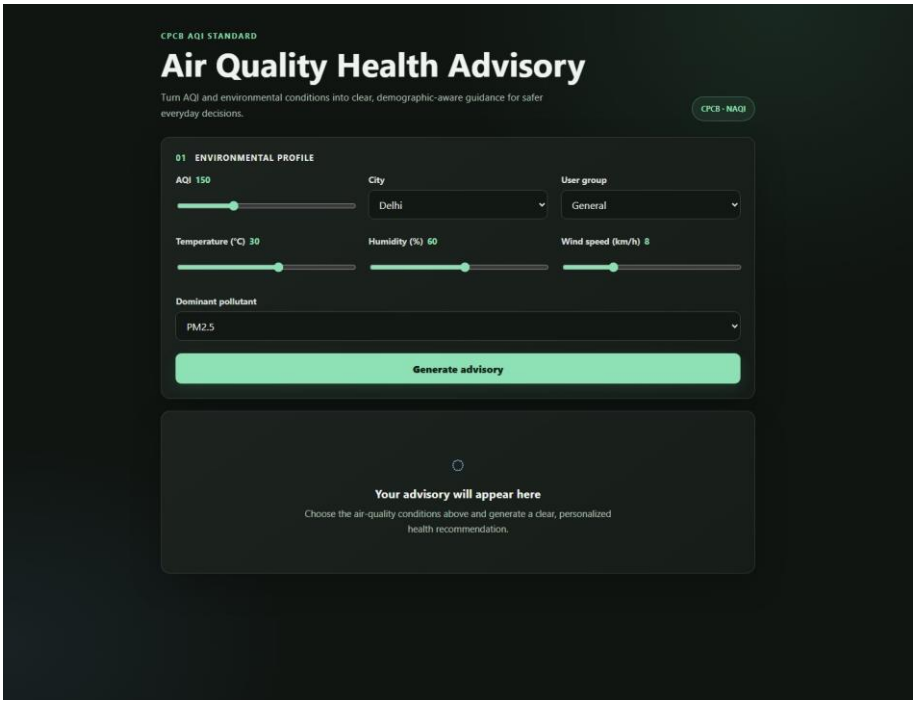
10. DASHBOARD DESIGN AND USAGE

The planned user interface is a Streamlit web dashboard. The user selects a city and target group, enters or adjusts AQI and weather values, chooses the main pollutant, and activates the GO control to generate an advisory.

10.1 Intended workflow

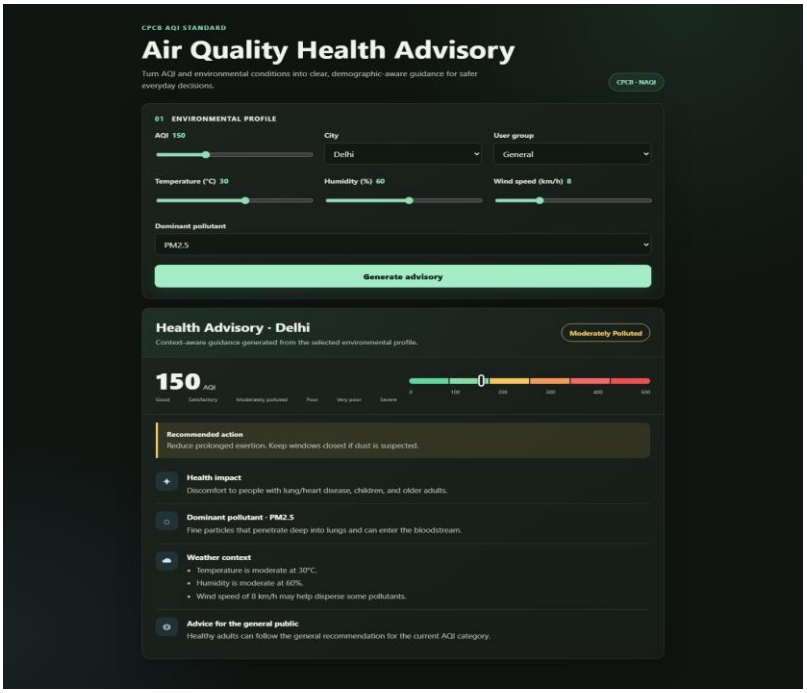
10. Start the application from the project environment using: streamlit run app/dashboard.py
11. Open the local address displayed by Streamlit, expected to be http://localhost:8501.
12. Select one of the supported cities: Bengaluru, Chennai, Delhi, Hyderabad or Mumbai.
13. Enter or confirm AQI and pollutant values, then select the main pollutant.
14. Adjust temperature, humidity and wind information when available.
15. Choose the target population group.
16. Select GO to display the risk category, health impact and personalized advisory.

The command and local address above are supplied in the project brief. Successful execution depends on the application files, Python environment, installed dependencies and availability of the required data.



10.2 Suggested dashboard output fields

- Selected city and observation time.
- Observed AQI and CPCB category.
- Relevant pollutant measurements and dominant pollutant.
- Weather context, where available.
- Forecast concentration and prediction horizon, if the forecast module is enabled.
- Health-impact summary and group-specific recommendations.
- A note about missing data, uncertainty or official alerts when relevant.



11. MODEL PERFORMANCE AND EVALUATION

11.1 Reported machine-learning performance

Model	Accuracy supplied
LightGBM	Approximately 89.9%
XGBoost	Approximately 89.8%
Random Forest	Approximately 87.7%

These values are taken from the project information provided and are not independently verified in this report. The final report should specify what label is being predicted, the dataset size and source, preprocessing steps, train/validation/test split, class balance and whether the results were obtained from an untouched test set.

11.2 Forecasting evaluation

For the LSTM pollutant-forecasting component, MAE and RMSE can quantify the difference between predicted and observed pollutant concentrations. Results should be reported separately for each pollutant and forecast horizon when feasible. A chronological split is recommended for time-series data.

11.3 Advisory evaluation

Relevance: does the advisory reflect the measured or forecast pollution condition?

Readability: can a non-technical user understand the message without specialist knowledge?

Accuracy: are the category and health recommendations supported by the defined rules and approved sources?

Consistency: do similar inputs produce compatible advice?

Personalization: does the wording address the selected user group appropriately?

Safety: does the output avoid diagnosis, unsupported medical claims and contradictions with official alerts?

A human review checklist should be used alongside automatic checks. Text-generation quality should not be evaluated only by fluency; factual alignment with the structured inputs is essential.

12. EXPECTED OUTCOMES

- A working Streamlit dashboard for exploring air-quality conditions across the selected Indian cities.
- CPCB-aligned classification into six air-quality categories.
- An LSTM-based component for forecasting pollutant concentrations over time.
- A Transformer-based NLP module that converts structured risk information into clear health advisories.
- Personalized recommendations for children, elderly people, sensitive individuals, people with asthma and general users.
- Improved readability and usefulness compared with a numerical-only display.
- A documented evaluation of prediction quality and advisory relevance, readability, consistency and safety.

13. LIMITATIONS AND SAFETY CONSIDERATIONS

Data availability: monitoring coverage and pollutant measurements may vary by city, station and time. Missing data must be identified rather than treated as clean air.

Forecast uncertainty: future pollutant concentrations are estimates and can be affected by sudden emissions, weather changes and sensor issues.

Model generalization: models trained on one city or season may not perform equally well in other locations or periods.

Language-model errors: a Transformer-based model may produce fluent but unsupported statements. Structured prompts, approved recommendation rules and post-generation checks are required.

Health limitations: the system provides general public-health information and is not a medical diagnosis or a substitute for professional advice.

CPCB consistency: the exact official AQI breakpoints, category names, pollutant averaging periods and health messages must be checked against current CPCB documentation before deployment.

Privacy and accessibility: the dashboard should avoid collecting unnecessary personal health information and should use plain language that is accessible to a broad range of users.

14. FUTURE ENHANCEMENTS

- Integrate reliable, regularly updated air-quality and weather data sources.
- Add timestamps, station information and data-quality indicators to improve transparency.
- Evaluate LSTM forecasts across multiple pollutants, cities and prediction horizons.
- Compare the LSTM against simple baselines to establish whether the added model complexity is useful.
- Use retrieval from approved CPCB and public-health guidance to ground generated advisories.
- Add multilingual advisory generation for Indian users while preserving the meaning of official guidance.
- Provide uncertainty indicators and clear separation between observed data and forecasts.
- Conduct usability testing with representative users and expert review of health messages.
- Monitor the models after deployment and periodically re-evaluate their performance.

15. CONCLUSION

The Air Quality Health Advisory Generator aims to make air-quality information more understandable and actionable by combining CPCB-based risk classification, machine-learning methods, LSTM pollutant forecasting and Transformer-based natural-language generation. Instead of presenting only AQI and pollutant values, the system is designed to explain the likely health impact and provide practical guidance suited to the selected population group.

The proposed Streamlit dashboard brings these functions together through a simple interface for selecting a city, entering air-quality and weather information, choosing a pollutant and user group, and generating an advisory. The reported LightGBM, XGBoost and Random Forest accuracy figures provide an initial performance reference, but they must be supported by documented experimental settings in the final

implementation. The reliability of the project will depend on correct CPCB thresholds, suitable data, careful forecasting evaluation and strict checks on generated text.

Overall, the project demonstrates how NLP can serve as a communication layer between environmental measurements and the public. With validated data, transparent uncertainty and expert-reviewed recommendations, it has the potential to support better awareness of air pollution and more informed exposure-reduction decisions.

REFERENCES

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APPENDIX A: IMPLEMENTATION CHECKLIST

- Confirm official CPCB category names, breakpoints and pollutant-specific rules.
- Record dataset sources, dates, units, missing-value handling and city coverage.
- Document the target variable and train/test procedure for LightGBM, XGBoost and Random Forest.
- Evaluate LSTM forecasts on chronological holdout data and report MAE/RMSE.
- Constrain the language model to approved health guidance and validated input fields.
- Test cases at each CPCB category boundary and for missing or contradictory inputs.
- Review sample outputs for every supported user group and city.
- Clearly label observed values, model forecasts and generated recommendations in the interface.